

Original Article

Assessment of Groundwater Quality Using Water Quality Index (WQI) and Linear Regression Analysis for Rural Areas of Latur

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Abstract – Groundwater is one of the most important sources of freshwater used for drinking, agriculture, and domestic purposes in rural areas. However, increasing population, agricultural activities, and improper waste disposal have led to the deterioration of groundwater quality. This study evaluates groundwater quality in rural areas of Latur using the Water Quality Index (WQI) method and statistical tools such as correlation and linear regression analysis. Various physicochemical parameters including pH, electrical conductivity, turbidity, total dissolved solids (TDS), chloride, sulphate, fluoride, iron, total hardness, calcium, magnesium, and alkalinity were analyzed. Groundwater samples were collected during pre-monsoon and post-monsoon seasons and tested using standard laboratory methods. The WQI approach was used to determine the suitability of groundwater for drinking purposes, while regression analysis was applied to identify relationships between water quality parameters. The results provide valuable insights into groundwater quality status and assist in developing strategies for sustainable groundwater management in rural regions.

Keywords – *Groundwater Quality, Water Quality Index (WQI), Linear Regression, Correlation Analysis, Rural Water Resources.*

1. INTRODUCTION:

Freshwater is one of the most essential natural resources for sustaining human life and maintaining ecological balance. A significant portion of the Earth's freshwater is stored in glaciers and ice caps, leaving only a small fraction available for direct human use. Among the available freshwater resources, groundwater accounts for more than 90% of accessible freshwater and plays a critical role in meeting drinking water and agricultural needs.

Groundwater is generally considered safer than surface water because it undergoes natural filtration through soil and rock formations. However, contamination from agricultural fertilizers, industrial waste, and domestic sewage can significantly affect groundwater quality. Poor water quality may

lead to various health issues and environmental problems.

Assessment of groundwater quality requires the analysis of several physicochemical parameters such as pH, turbidity, electrical conductivity, total dissolved solids (TDS), chloride, sulphate, fluoride, iron, hardness, calcium, magnesium, and alkalinity. The Water Quality Index (WQI) is widely used to simplify complex water quality data and provide a single value representing overall water quality status. Statistical techniques such as correlation and regression analysis are also useful in understanding relationships among water quality parameters and predicting groundwater quality trends.

2. STUDY AREA:

The study was conducted in rural areas of Latur district located in the state of Maharashtra, India. The region lies within the Deccan plateau and is predominantly covered by basaltic lava flows known as Deccan Trap formations.

The climate of Latur is semi-arid with hot summers and moderate rainfall during the monsoon season. Groundwater is the primary source of drinking water for rural communities, mainly extracted through dug wells and bore wells.

A. Hydrogeology: The geology of the study area consists mainly of basaltic rocks formed during volcanic eruptions in the Deccan Trap period. These formations are composed of multiple lava layers that create plateau-like terrain. The basalt formations can be broadly classified into two types.

- Massive basalt, which is relatively impermeable.
- Vesicular basalt, which contains pores and cavities capable of storing groundwater.

Weathering and fracturing of these basalt layers significantly enhance groundwater storage capacity.

B. Aquifer Characteristics: Groundwater in the region occurs in different aquifer zones:

- Shallow aquifer: up to approximately 20 m depth, consisting of weathered basalt
- Intermediate aquifer: between 20 m and 40 m depth, consisting of fractured basalt
- Deep aquifer: below 40 m depth with confined groundwater conditions

The availability of groundwater depends on factors such as fracture density, weathering intensity, and geological structures.

3. WATER QUALITY PARAMETERS:

Groundwater quality is evaluated using several physicochemical parameters.

- A. pH: The pH value indicates whether water is acidic or alkaline. Drinking water should typically have a pH between 6.5 and 8.5.
- B. Electrical Conductivity: Electrical conductivity measures the ability of water to conduct electricity and indicates the concentration of dissolved ions.
- C. Total Dissolved Solids (TDS): TDS represents the total concentration of dissolved salts and minerals in water.
- D. Total Hardness: Hardness is mainly caused by dissolved calcium and magnesium ions.
- E. Chloride: High chloride concentrations may give water a salty taste and may contribute to corrosion.
- F. Alkalinity: Alkalinity represents the capacity of water to neutralize acids.
- G. Fluoride: Fluoride in small amounts is beneficial for dental health, but excessive fluoride can cause fluorosis.
- H. Calcium and Magnesium: These minerals contribute to hardness and scaling in water systems.

4. METHODOLOGY:

- A. Sample Collection: Groundwater samples were collected from various wells and bore wells in rural areas of Latur district during both pre-monsoon and post-monsoon seasons.
- B. Laboratory Analysis: The collected samples were analyzed for various physicochemical parameters using standard laboratory methods recommended for drinking water analysis.
- C. Water Quality Index (WQI): The Water Quality Index method was used to evaluate overall groundwater quality. The WQI combines multiple water quality parameters into a single numerical value representing the overall quality of water.

Steps involved in WQI calculation include:

- Selection of water quality parameters
- Assignment of weights to each parameter
- Calculation of relative weights
- Determination of quality rating scale
- Calculation of overall WQI value

Based on WQI values, water quality is classified into categories such as:

WQI Range	Water Quality
0–50	Excellent
50–100	Good
100–200	Poor
200–300	Very Poor
>300	Unsuitable for Drinking

D. Correlation Analysis: Pearson’s correlation coefficient was used to determine the relationships among different water quality parameters. This analysis helps in identifying parameters that significantly influence groundwater quality.

E. Linear Regression Analysis: Linear regression analysis was applied to develop predictive relationships between highly correlated parameters. This statistical method helps estimate the value of one parameter based on another parameter.

5. RESULTS AND DISCUSSION:

The results obtained from laboratory analysis indicate variations in groundwater quality across different sampling locations. Seasonal changes also influenced the concentration of various parameters.

Higher values of TDS, hardness, and chloride were observed in certain locations, indicating mineral dissolution and possible anthropogenic influences.

Correlation analysis revealed strong relationships between several parameters such as:

- Electrical conductivity and TDS
- Hardness and calcium
- Hardness and magnesium

Regression models developed in this study can be used to estimate water quality parameters with reduced laboratory testing.

The calculated WQI values indicate that groundwater in some areas is suitable for drinking, while other locations require treatment before consumption.

6. CONCLUSION:

Groundwater quality assessment in rural areas of Latur was carried out using the Water Quality Index method along with correlation and regression analysis. The study revealed that groundwater

quality varies across locations and seasons due to geological and anthropogenic factors.

The WQI results indicate that while some groundwater sources are suitable for drinking, others require treatment due to higher concentrations of dissolved minerals. Statistical analysis showed strong relationships between several water quality parameters, which can be useful for predicting groundwater quality trends.

Regular monitoring and proper groundwater management strategies are necessary to ensure safe and sustainable water supply in rural regions.

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